

CSCE 636 DEEP LEARNING · MODULE 10 · LECTURE NOTES

Generative Models

Learning a distribution instead of a mapping.

MODULE OUTCOMES

Distinguish the generative model families by what they optimise.

Explain VAEs and the reparameterisation trick.

Explain GANs and why they are hard to train.

Explain diffusion models and why they won.

Explain classifier-free guidance and conditioning.

1 The problem

GENERATIVE MODELLING LEARNS $P(X)$, NOT $P(Y|X)$

A discriminative model learns a mapping from inputs to outputs. A generative model learns the *distribution* the data was drawn from, well enough to produce new samples from it.

That is a substantially harder problem. A classifier needs only to represent what distinguishes the classes and may ignore everything else; a generative model must capture everything about the data, including all the structure that no label ever referred to.

And evaluation is genuinely hard. There is no held-out accuracy. **'Is this a good sample?'** resists **measurement**, the standard metrics are known to be flawed (§4), and the thing you actually want — samples a person would judge good and varied — has no agreed formalisation.

The families differ in how they make the problem tractable, and each makes a different compromise among sample quality, sample diversity, sampling speed, and whether the likelihood can be computed at all.

Family	What it optimises	The trade
Autoregressive	The exact likelihood, factorised as a product of conditionals.	Excellent quality and a tractable likelihood; sampling is sequential and therefore slow, element by element. The dominant approach for text (Module 07).
Variational autoencoder	A lower bound on the likelihood (the ELBO).	Fast to sample, stable to train, and blurry — for the reason in §2.

Family	What it optimises	The trade
GAN	A minimax game between a generator and a discriminator. No likelihood at all.	Sharp samples; notoriously unstable training; mode collapse, where the generator produces a few outputs that fool the discriminator and abandons the rest of the distribution.
Diffusion	Denoising at many noise levels — a simple regression objective.	Best sample quality, stable training, slow sampling (many network evaluations), which distillation addresses.
Normalising flows	Exact likelihood via a sequence of invertible transformations.	Exact and elegant; the invertibility requirement heavily constrains the architecture, which limits capacity.

Diffusion won image generation because it offered GAN-level sample quality with VAE-level training stability, which nothing else did. That single sentence explains the field's shift around 2021 — GANs had the quality and were miserable to train; VAEs trained easily and produced blur; diffusion took the good half of each.

2 Variational autoencoders

THE REPARAMETERISATION TRICK MAKES SAMPLING DIFFERENTIABLE

A VAE encodes an input to a *distribution* over latent variables — a mean and a variance — samples a latent from it, and decodes that sample back to the input space. The sampling step is the problem: you cannot backpropagate through a random draw, because the draw is not a differentiable function of its parameters.

So move the randomness outside the computation. Instead of sampling $z \sim N(\mu, \sigma^2)$ directly, sample $\epsilon \sim N(0, 1)$ and compute $z = \mu + \sigma \cdot \epsilon$.

Now ϵ is an *input* to the graph rather than an operation within it, and the gradient flows through μ and σ by ordinary arithmetic — Module 02's machinery applies without modification.

That single restructuring is what made latent variable models trainable by gradient descent, and the same trick appears throughout machine learning wherever a stochastic node must be differentiated through — in reinforcement learning, in variational inference generally, and in the Gumbel-softmax relaxation for discrete choices. It is one of the most reusable ideas in this course.

WHY VAE SAMPLES ARE BLURRY

The reconstruction term is typically squared error, which is maximum likelihood under an assumption of Gaussian output noise (CSCE 633 Module 10 §1) — an assumption nobody examined and which is doing a great deal of work.

Squared error is minimised by the *mean* of the plausible outputs. If several different sharp images are equally consistent with a given latent code, the loss is lowest at their average rather than at any one of them.

And the average of several sharp images is a blurry image. The model is not failing to represent sharpness; it is correctly declining to commit.

So the blur is the loss function working exactly as specified, not a capacity failure — **which is why adding parameters does not fix it and changing the reconstruction loss does.** Perceptual losses, adversarial losses, and discrete latent spaces (VQ-VAE) are all attacks on this one issue, and the VQ-VAE route — a discrete latent plus an autoregressive prior — became the basis of several strong systems.

3 Diffusion

FORWARD PROCESS -- fixed, nothing learned
 repeatedly add a little Gaussian noise over T steps until the image is indistinguishable from pure noise.
 Closed form: you can jump directly to any step t .

REVERSE PROCESS -- learned
 train a network to predict the NOISE that was added at step t , given the noisy image and t .
 Loss: plain mean squared error on the predicted noise.

SAMPLING
 start from pure noise and apply the denoiser repeatedly, stepping t down to 0.

WHY IT WORKS SO WELL

- * the training objective is a STABLE regression -- no adversarial game and no balance to maintain
- * each step is an EASY problem (remove a little noise); one hard problem decomposed into many easy ones
- * the model sees every noise level, so it learns coarse structure (high noise) and fine detail (low noise)

COST: sampling needs many network evaluations. DDIM and distillation take 1000 steps down to 20, or to 1.

CLASSIFIER-FREE GUIDANCE CONTROLS QUALITY AGAINST DIVERSITY

Train a single model both conditionally and unconditionally, by randomly dropping the conditioning signal (the text embedding, the class label) on some fraction of training examples.

At sampling time, extrapolate away from the unconditional prediction toward the conditional one: the guided prediction is the unconditional one plus a scale factor times the difference. **A scale of 1 is ordinary conditional sampling; higher scales exaggerate the conditioning.**

High guidance produces samples that adhere closely to the prompt and are less diverse — and eventually over-saturated and artefact-laden. **Low guidance gives diversity and weaker adherence.**

So a single knob trades prompt fidelity against variety, at sampling time, with no retraining — which is most of why text-to-image systems feel controllable, and it is a rare case of a technique that is both simple to implement and decisive for the user experience.

4 Evaluation

- **Fréchet Inception Distance (FID).** The distance between Gaussian fits to the Inception features of real and generated sets. **The standard metric, and it is sensitive to the sample count, the resizing method, and the exact Inception implementation** — published FIDs are frequently not comparable across papers, which is a measurement problem of exactly the kind CSCE 633 Module 12 described.

- **Inception Score.** Older, and it **cannot detect mode collapse within a class** — a model producing one perfect image per class scores well.
- **Precision and recall for generative models.** Separates *quality* (are the samples realistic?) from *coverage* (do they span the distribution?), **which a single number cannot** — and the two fail in different ways, so reporting both is far more informative.
- **Human evaluation.** Expensive, slow, and **still the only thing that measures what is actually wanted** — with all the inter-annotator agreement caveats of CSCE 633 Module 11 §4.
- **And likelihood, where the family provides it** — which **correlates surprisingly poorly with perceived sample quality**, a known and somewhat embarrassing result for the likelihood-based families. **Report samples, including failures:** in this area a metric is a summary of something nobody has defined precisely, and the pictures carry more information than the number.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Lilian Weng — What are Diffusion Models? (free)	The clearest derivation of §3 available , including the connection to score matching.
Kingma & Welling — Auto-Encoding Variational Bayes (free)	The §2 reparameterisation trick , from the paper that introduced it.
Ho, Jain & Abbeel — Denoising Diffusion Probabilistic Models (free)	The §3 formulation that made diffusion practical, with the simplified objective.
Ho & Salimans — Classifier-Free Diffusion Guidance (free)	The §3 guidance mechanism. Two pages of idea and it is behind every text-to-image system.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement a VAE on a small image dataset and **sample from the prior**.
- 2 **Implement sampling without the reparameterisation trick** and show the gradient does not flow.
- 3 **Demonstrate the blur mechanism:** average several sharp images and compare against your VAE's samples.
- 4 Replace the squared-error reconstruction with a perceptual loss and compare sharpness.
- 5 **Implement a diffusion model** on small images, with the simple noise-prediction objective.

- 6 **Visualise the reverse process** — show the sample at several steps from noise to image.
- 7 Implement DDIM sampling and plot quality against step count.
- 8 **Implement classifier-free guidance** and sweep the guidance scale. Show the quality–diversity trade.
- 9 **Compute FID with two different implementations** and report whether they agree.
- 10 Compute precision and recall for your generative model and show they say different things.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 How does generative modelling differ from discriminative, and why is evaluation hard?
- 2 Compare five generative families on what each optimises.
- 3 Why did diffusion win image generation?
- 4 State the reparameterisation trick and say what problem it solves.
- 5 Why are VAE samples blurry, and what fixes it?
- 6 Describe the diffusion forward and reverse processes.
- 7 Give three reasons diffusion trains stably.
- 8 Explain classifier-free guidance and what it trades.
- 9 Give five generative metrics and the problem with each.